METRIC

MIL-PRF-24623/5C
w/AMENDMENT 2
2 September 2026

SUPERSEDING
MIL-PRF-24623/5C
w/AMENDMENT 1
23 May 2022

PERFORMANCE SPECIFICATION SHEET

SPLICE, FIBER OPTIC, CABLE, HOUSING

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

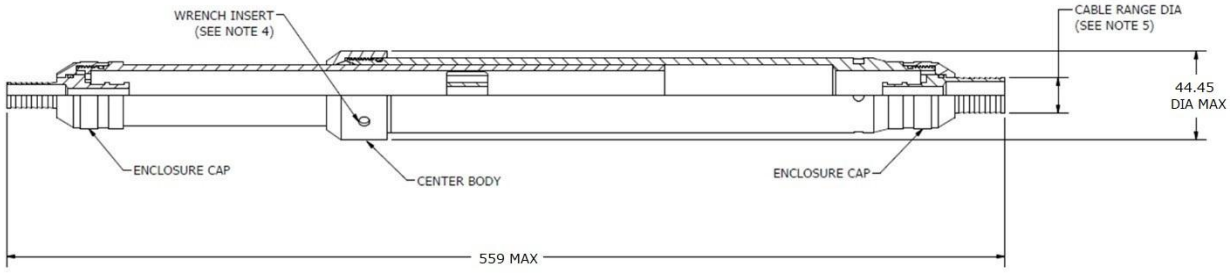
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-24623.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 6060

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:27Z
Check the source to verify that this is the current version before use.



Millimeters	Inches
559	22.00
44.45	1.75

NOTES:

1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Finish to be non-conductive.
4. Adjustable pin spanner wrench insert, 3⁄16-inch depth maximum.
5. Cable splice housing for four-fiber/sub-unit cables shall accommodate a cable range from 7.75 to 8.50 millimeters (0.305 to 0.344 inch). Cable splice housing for eight-fiber/sub-unit cables shall accommodate a cable range from 10.75 to 11.50 millimeters (0.423 to 0.453 inch).

FIGURE 1. Cable splice housing.

REQUIREMENTS:

Design and construction: See figure 1. The cable splice housing shall be constructed such that relative rotation of center body and backshell/backnut assembly is prevented. Cable splice housing shall include an internal splice nest to house the specified number of splices. A splice nest for M24623/5-01 and M24623/5-02 shall be constructed to interface with the single fiber fusion splice protector in accordance with MIL-PRF-24623/6. A splice nest for M24623/5-03 and M24623/5-04 shall be constructed to interface with a ribbon fiber splice protector in accordance with MIL-PRF-24623/8.

Finish: Plating of the cable splice housing shall be a non-conductive coating in accordance with MIL-PRF-8625, type III, class 2, sealed or unsealed, with supplemental impregnated, polytetrafluoroethylene (PTFE) coating. Supplemental PTFE coating shall be the result of impregnating PTFE into the surface asperities of the anodic coating. Buildup per surface for this non-conductive finish shall be between 0.018 and 0.031 millimeter (0.0007 and 0.0012 inch). Color of the non-conductive finish shall be black in accordance with EIA-359, and the color shall be minimally reflective.

Fiber style: Single mode or multimode.

Splice type: Cable splice housing.

Weight: Less than or equal to 550 grams.

Temperature range: 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:27Z
Check the source to verify that this is the current version before use.

Materials:

Housing: Aluminum alloy in accordance with ASTM B211/B211M recommended.

O-rings: Fluoro-silicone material recommended.

Splice nest: Fluoro-silicone material is recommended for a resilient splice nest. Aluminum is recommended for a rigid splice nest.

VERIFICATION:

Verification shall be performed using single mode cable.

Nuclear radiation: Not applicable.

Insertion loss: 0.20 decibel per fiber splice.

Return loss: Not applicable.

Crosstalk: Not applicable.

Ambient light susceptibility: Not applicable.

Fiber pull-out force: Not applicable.

Vibration: Applicable.

Shock: Applicable.

Part or Identifying Number (PIN):

M24623/5-01 cable splice housing for eight-fiber cable (see MIL-PRF-85045/17).

M24623/5-02 cable splice housing for four-fiber cable (see MIL-PRF-85045/18).

M24623/5-03 cable splice housing for eight sub-unit cable (see MIL-PRF-85045/35).

M24623/5-04 cable splice housing for four sub-unit cable (see MIL-PRF-85045/34).

QUALIFICATION: With respect to products requiring qualification, awards will be made only for products, which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24623 whether or not such products have actually been solicited by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Issac Hull Avenue, SE, Step 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

Qualification by similarity (M24623/5-02): Manufacturers that produce products for M24623/5-01 and are qualified under M24623/5-01 and produce products to this specification sheet for M24623/5-02 using the same materials and processes as the qualified product (M24623/5-01) and meet the size inspection, workmanship, cable retention, cable seal flexing, and twist requirements specified in MIL-PRF-24623 are qualified under this specification sheet for M24623/5-02.

Qualification by similarity (M24623/5-04): Manufacturers that produce products for M24623/5-03 and are qualified under M24623/5-03 and produce products to this specification sheet for M24623/5-04 using the same materials and processes as the qualified product (M24623/5-03) and meet the size inspection, workmanship, cable retention, cable seal flexing, and twist requirements specified in MIL-PRF-24623 are qualified under this specification sheet for M24623/5-04.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:27Z
Check the source to verify that this is the current version before use.

Qualification retention: As long as the material supplied is identical in every respect to the qualification sample tested and is found satisfactory, then the period for group C inspections for qualification retention is to be every 5 years.

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
- Army – CR
- Navy – SH
- Air Force – 85
- NASA – NA

Preparing activity:
- Navy – SH
- (Project 6060-2026-004)

Review activities:
- Army – AR, MI
- Navy – AS, CG, EC, MC
- Air Force – 19, 70
- DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:27Z
Check the source to verify that this is the current version before use.